

Melkyn Quintana

📍 Medellín - Colombia ✉ melkyn.quintana@gmail.com ☎ +57 3219964364 🌐 My Website

Professional Profile

I am an astronomer with a strong background in mathematics, physics, statistics, data analysis, and Python/C programming. I'm passionate about cosmology, black holes, and everything that helps us understand the evolutionary history of the universe. In fact, my research work for my graduation thesis focuses on constraining cosmological parameters through the comparison and statistical analysis of SNe Ia data. Throughout my academic and professional career, I have acquired strong learning and problem-solving skills, which allow me to easily adapt to new work and collaborative environments.

Education

B. Sc. in Astronomy 2019 – 2026

University of Antioquia

- **Courses:** Computational Methods, Modern cosmology, Modern astrophysics, Stellar astrophysics, Galactic and extragalactic astrophysics, Celestial mechanics, Astro-statistics, Quantum Mechanics, Relativity and Gravitation.

Technician in Industrial Instrumentation 2017 – 2018

Servicio Nacional de Aprendizaje (SENA)

Technical High School with emphasis on electronics 2013 – 2018

Technic and Industrial Institute Rafael Reyes

Work Experience

Bilingual Costumer Service Agent October 2024 - Current

iQor

- Dedicated Customer Service Agent with experience supporting the Walmart Spark delivery network, specializing in logistics coordination, driver assistance, and conflict resolution

Freelancer December 2024 -

Outlier

December 2025

- Contribute to projects focused on developing and optimizing prompts, reviewing code, and evaluating responses for Large Language Models (LLM) training. Participate in voice recording initiatives to improve model accuracy and performance.

3D Image Annotator January 2025 - May 2025

CloudFactory

- Conducted detailed roof modeling and annotation using Nearmap's 3D satellite imagery platform. Utilized CAD-like tools to trace and refine roof structures, ensuring accurate measurements and improved visual clarity by correcting distortions in 3D renderings caused by image resolution limitations.

Tech Skills

Operating Systems: GNU/Linux, Windows

Programming Languages: Python, C, Bash, SQL, ssh, JavaScript

Software: Git, L^AT_EX, Iraf, Jupyter

Tools: Statistics, Bayesian Inference, Monte Carlo methods, emcee, ML, Big Data analysis, pandas, Numeric methods, N-Body Simulations, TensorFlow, React, Power BI.

Courses and Certificates

BOOTCAMP TALENTO TECH - DATA ANALYSIS ADVANCED LEVEL 2025
(159 hrs)

Ministry of Information and Communications Technology of Colombia / UI Training

DATA - DRIVEN ASTRONOMY (20 hrs) 2022
Coursera / University of Sydney

SPECIALIZED PROGRAM - PYTHON FOR EVERYBODY (80 hrs) 2022
Coursera / University of Michigan

This specialized program is composed of 5 courses:

- Programming for Everybody (Getting Started with Python)
- Python Data Structures
- Using Python to Access Web Data
- Using Databases with Python
- Capstone: Retrieving, Processing, and Visualizing Data with Python

INTRODUCTION TO PYTHON PROGRAMMING (20 hrs) 2022
Coursera / University of Pennsylvania

DATA ANALYSIS USING PYTHON (20 hrs) 2022
Coursera / University of Pennsylvania

Languages

Spanish Native Speaker

English Fluent (B2 - B2+)